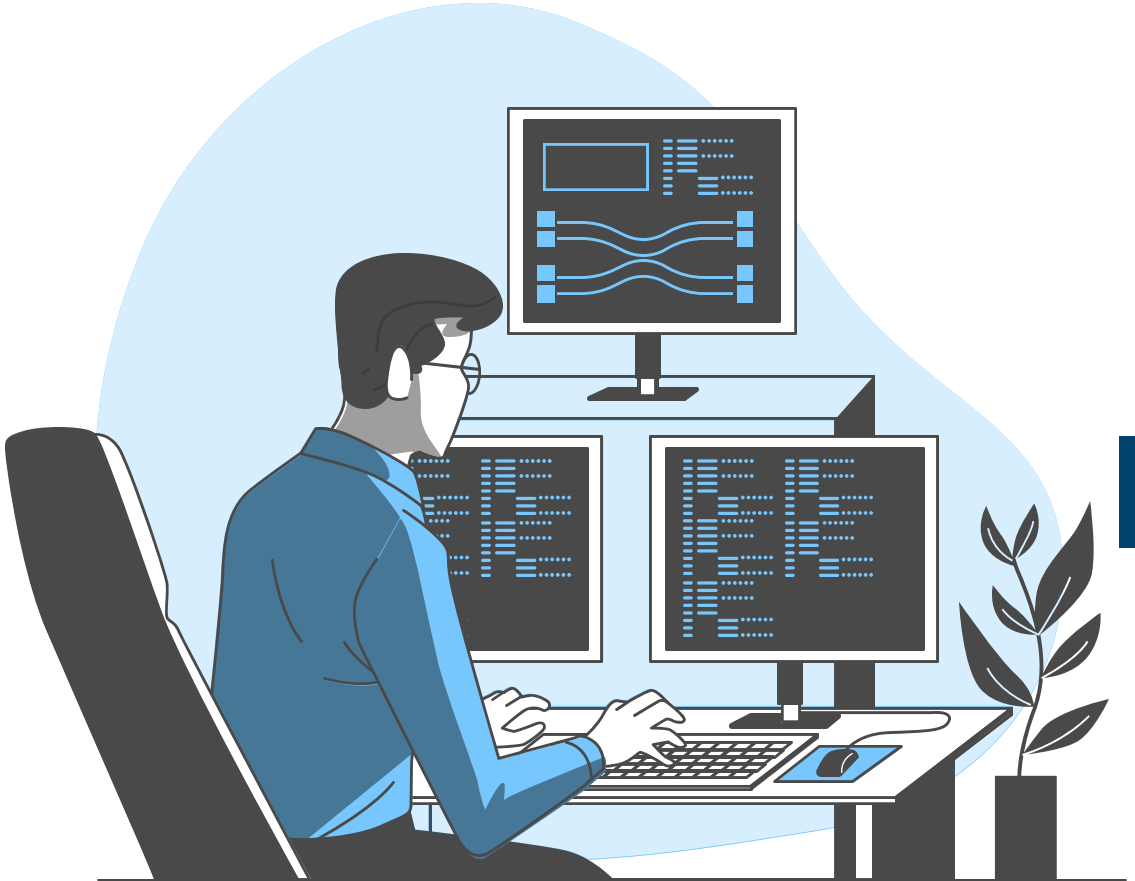

BIOTECH REINVENTED

A Comprehensive Industry Analysis

EXECUTIVE SUMMARY

This comprehensive report delves deeply into the intricate landscape of the biotech industry, meticulously examining its current challenges, promising opportunities, and future trajectories. With a keen focus on innovation, cutting-edge research, and evolving market trends, the analysis offers invaluable insights tailored for a diverse audience including stakeholders, investors, policymakers, and industry professionals.

Serving as a strategic compass, the report illuminates key facets of the biotech sector essential for well-informed decision-making. Through a rigorous synthesis of recent advancements, regulatory frameworks, and dynamic market forces, readers stand to gain a profound understanding of the industry's complexities.

Moreover, the report identifies and evaluates emerging technologies, assesses investment landscapes, and explores sustainable growth pathways. Covering pivotal areas such as Market Analysis, Technological Advancements, Regulatory Environment, Investment Landscape, and Future Outlook, this in-depth exploration provides readers with a holistic and nuanced perspective. Its aim is to empower readers with comprehensive knowledge, enabling them to navigate the multifaceted biotech landscape successfully.

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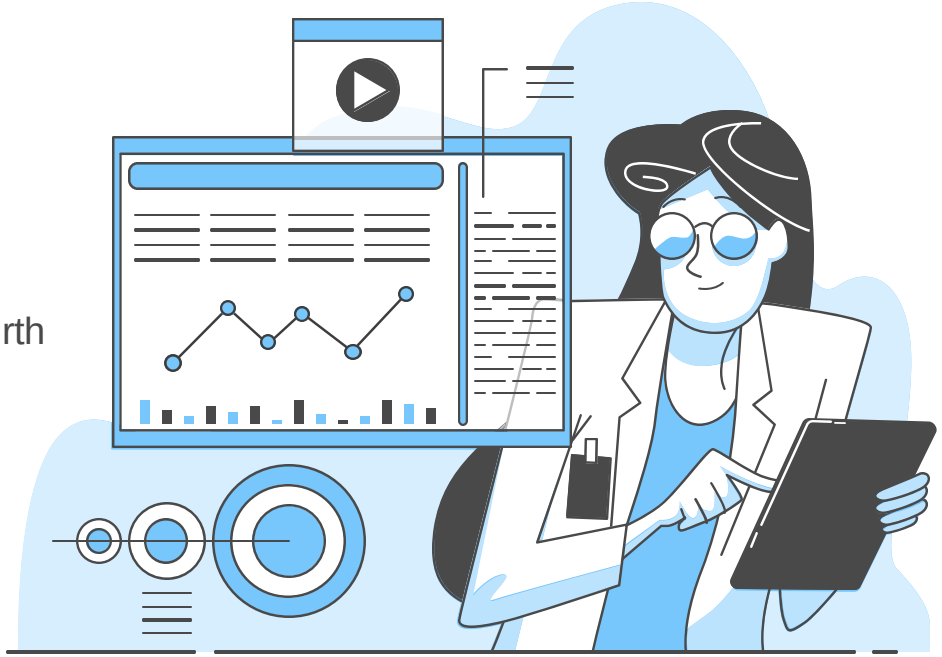
Definition, History, Market Size

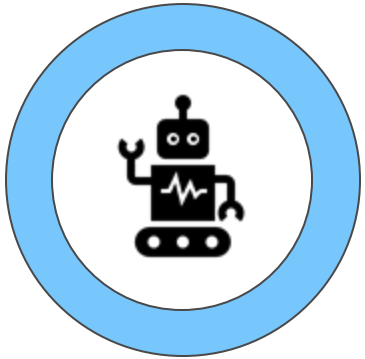
02 **Market Bifurcation**
Health, Agriculture and Natural Resources

03 **Regional Analysis**
Asia Pacific, Europe and North America

04 **Funding Landscape**
Problems, Market Trends, Our Opinion and Suggestions



BIOTECH OVERVIEW



INTRODUCTION

Biotechnology is the application of technology to use, change, or improve a component or the entire biological system for the benefit of industry and human welfare

HISTORY OF BIOTECH

01

In 1919, Hungarian engineer **Karl Ereky** used the term "**biotechnology**" to define a technology that involves turning raw materials into a more valuable product.

02

Ancient Egyptians used fermenting techniques based on a knowledge of **microbiological activities** that occur in the absence of oxygen.

03

Later studies revealed that yoghurt is created by the activity of **yeast** added to milk, which is likewise biotechnology because it employs a **microbe** for human benefit.

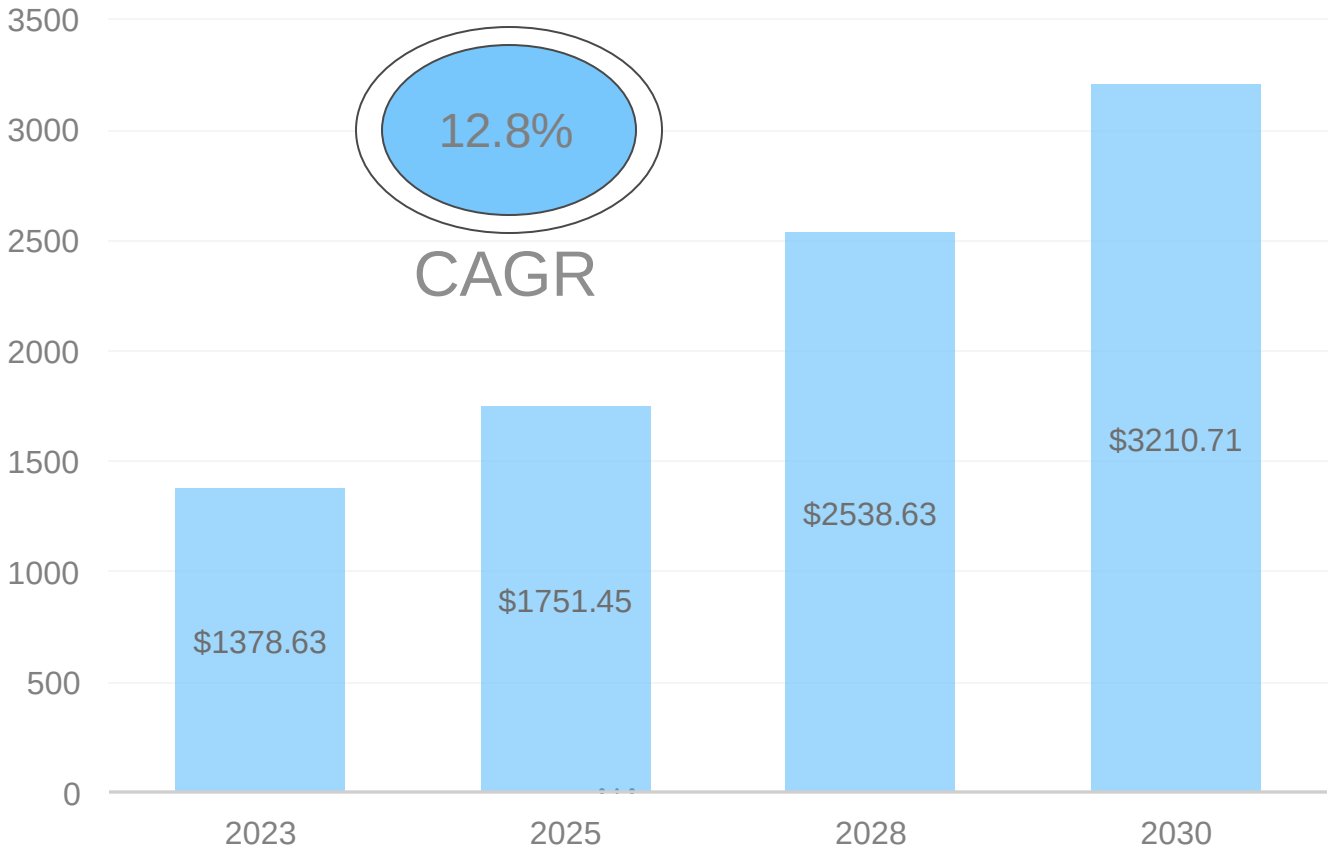
04

People have utilised **selective breeding** to enhance crop and livestock production in order to feed themselves. Organisms with **desired traits** are mated to generate offspring with the same qualities in selective breeding.

MARKET SIZE

The global biotechnology market was estimated at **USD 1,224.31** billion in 2022 and is expected to be worth around **USD 3,210.71** billion by 2030 and poised to grow at a noteworthy **CAGR** of **12.8%** from 2023 to 2030.

BIOTECH MARKET SIZE 2023 TO 2030



KEY TAKEAWAYS

01

North America accounted for **37.76%** revenue share in 2022.

02

Asia Pacific has generated a revenue share of **23.8%** in 2022.

03

By application, the **bio-pharmacy** segment has captured **41.46%** revenue share in 2022.

04

The **bio-industries** application segment held a **24.41%** revenue share in 2022.

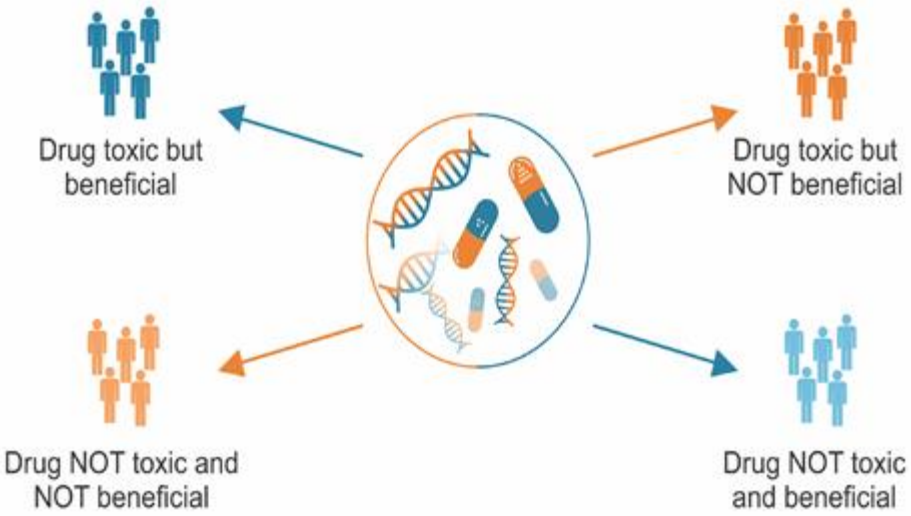
MARKET BIFURCATION

BIOTECHNOLOGY IN HEALTH

In the field of medicines, biotechnology is widely used in the development of several innovative techniques for diagnosing, treating and preventing diseases.

1. Pharmacogenomics

Pharmacogenomics is the study of the role of the genome in disease. It has led to the production of drugs that are best suited to an individuals genetic makeup. It can be applied in diseases such as Cancer, Depression, HIV, Asthama etc.



2. Edible Vaccines

Vaccines are obtained by animal and cell cultures. These vaccines contain inactivated pathogens which are used as follow:

- The transgenic plants can produce antigens that can be used as edible vaccines.
- Antigens proteins from several pathogens can be expressed in plants such as Tomato and Banana.
- Transgenic Sugarbeet can treat food and mouth disease of animals and can cure diseases such as cholera and hepatitis B.

3. Recombilant Insulin

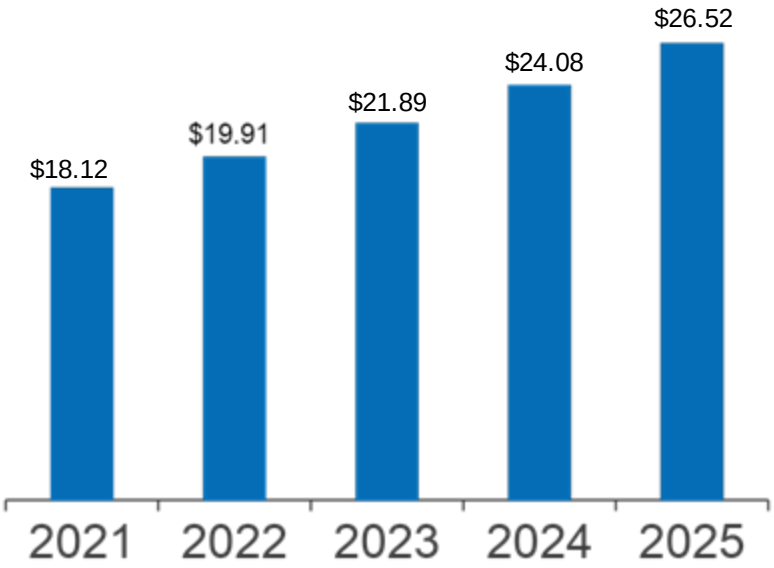
- Insulin is required by diabetic patients to remove excess sugar from the blood.
- Diabetic patients have very low or no insulin so they need external insulin.
- Later it was discovered that insulin produced by the pancreas of Pig can be used by humans but there were not enough pigs which led to the cloning of human insulin gene.

FOOD SECURITY AND SUSTAINABLE AGRICULTURAL DEVELOPMENT

- All the major countries are focusing on development of biotechnological strategies and biosafety frameworks to develop the capacities in agricultural biotechnologies and related issues through technical co-operation and training.

- Agricultural biotechnologies represent a large range of technologies that are used for various purposes.

- These are used for genetic improvement of plant varieties and animal populations, characterization and conservation of genetic resources , diagnosis of plant and animal diseases, vaccine development etc.



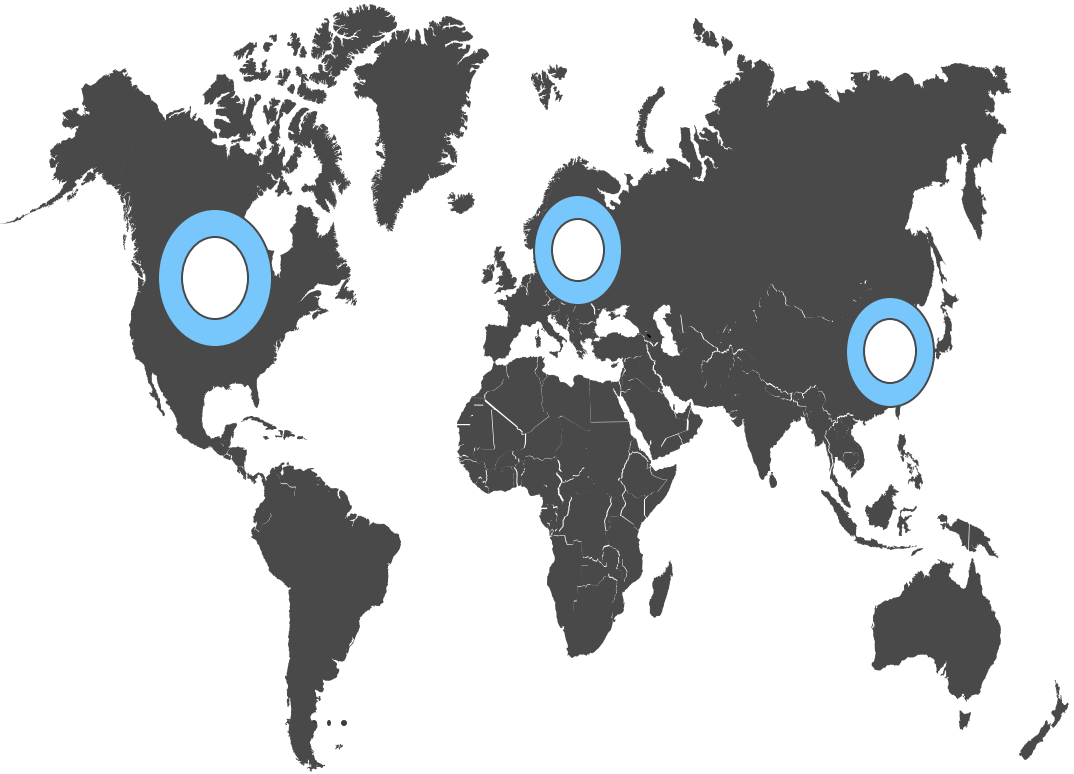
NATURAL RESOURCES

Biotechnology can be used for more than the creation of genetically engineered crops: it can be used to preserve biodiversity and aid ingenetic resource conservation efforts.

Agricultural Biotechnology can have secondary effects that enhance natural resources conservation and protect the environment and also conserve water resources.

There are various ways to do this like by developing drought-tolerant crops, conducting researches and using herbicide-resistant crops.

REGIONAL ANALYSIS



PUBLICATIONS BY EACH REGION

EUROPE

Europe, in particular, continues to be a scientific powerhouse: in 2021, it produced nearly **750,000** scientific publications.

ASIA-PACIFIC

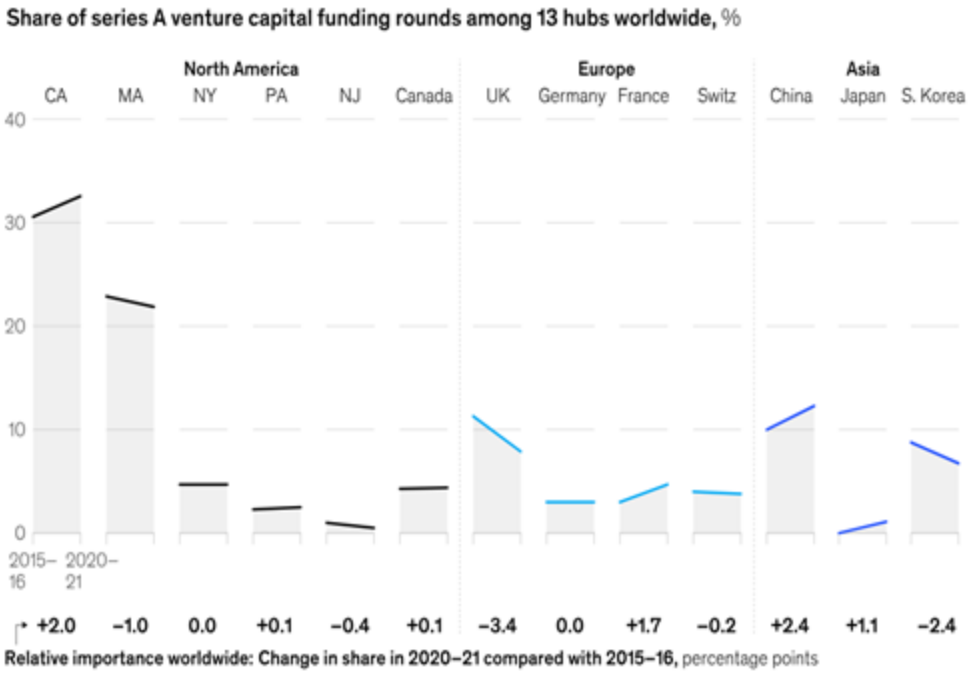
Up until the year 2020, the Asia-Pacific region contributed an estimated **950,000** research publications, indicating a substantial body of scholarly work.

NORTH AMERICA

In the field of scientific research, North America has been a prolific contributor, with an estimated **450,000** scientific publications produced up to the year 2021.

GLOBAL GROWTH OF PRIVATE CAPITAL

The availability of **private capital** continues to grow worldwide, suggesting that venture capitalists have considerable reserves to deploy. European healthcare-focused VC funds have grown at an annualized rate of 24 percent since 2012, though they raised only **$6 billion** in 2021, while their counterparts in the United States and China raised **$20 billion** and **$8.6 billion**, respectively.



FUNDING LANDSCAPE

Biotech has profoundly changed the sort of research Pharma conducts and the sort of products it makes but

HOW WELL HAS BIOTECH REALLY DONE?

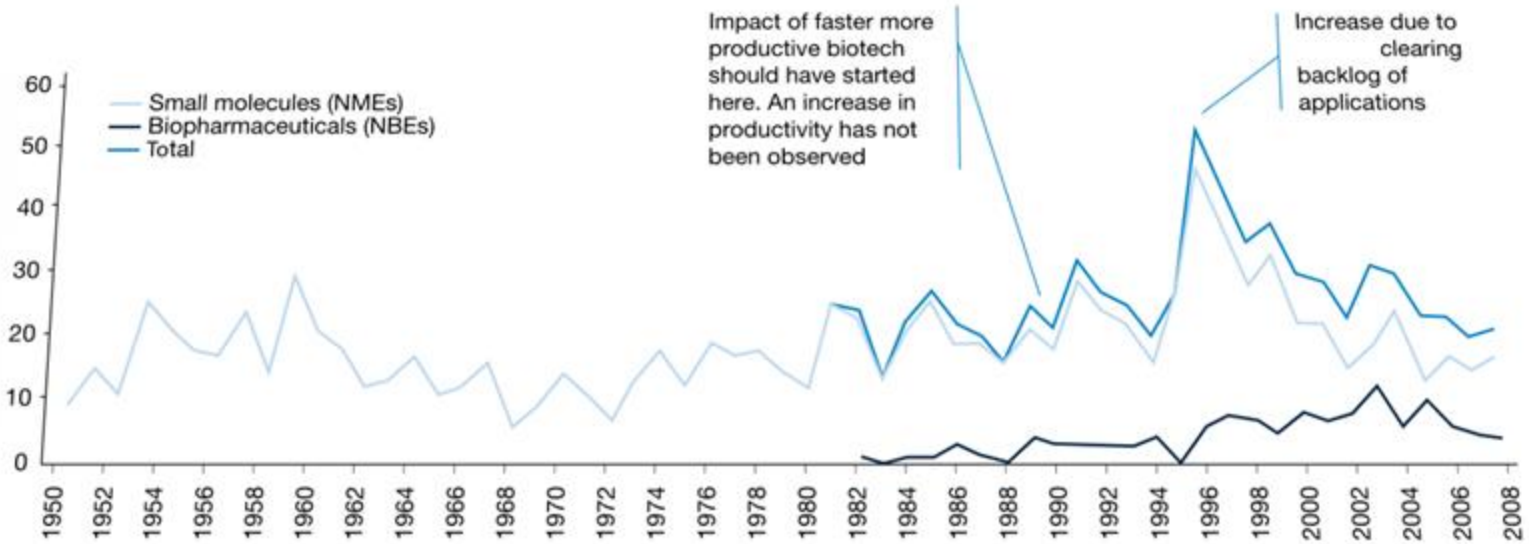
The good news is that it's produced some valuable new platform technologies and treatments. More than 100 different recombinant protein-based drugs and at least 40 'companion' diagnostics have also been launched, and some of these therapies have proved very effective in treating complex conditions

...

The bad news is that Biotech hasn't made a significant difference to Pharma's productivity. Between 1950 and 2008, the US FDA approved 1,222 therapies. Given that it takes about 10 years to develop a drug, the total number of approvals should have started rising in about 1990, if Biotech had succeeded in improving Pharma's output.

But, as the growth shows, the number of approvals has remained broadly constant.

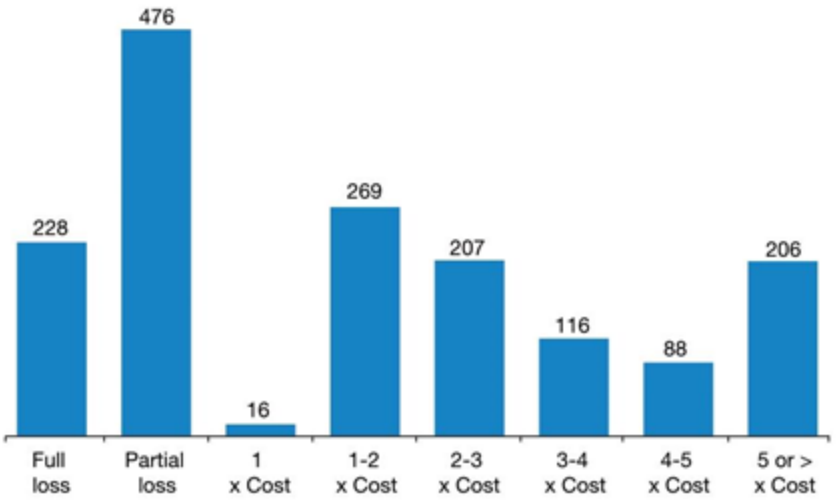
WHY SO?



The reason's simple: Biotech hasn't reduced the inherent risk in drug discovery and development.

A Business Model Standing on Pillars of Sand?

The business model on which Biotech has relied for the past 30 years is now breaking down. This model is based on external investment – typically, venture capital – in an innovative idea arising from an entrepreneurial source. It assumes that investors can realise value through one of two routes: flotation on the public markets or, more frequently, a trade sale to an established pharma company which carries a very high risk of failure. In one recent study of 1,606 biotech investments that were realised between 1986 and 2008, 704 investments resulted in a full or partial loss, while 16 only covered their costs.



What's Driving Such Trends?

Many of the external conditions that enabled these biotech companies to thrive are rapidly vanishing

01 The East Wind

The West, which has been the experiencing a 'reverse brain drain'. Numbers of students graduating in doctorates of physical and biological sciences in the East is at its all time high.

02 Capital Constraints

The recession has hindered biotech firms' capital raising, falling far below historical levels. Almost half of the funds were allocated to only a few established public companies in FPOs.

03 Hotter Competition

Emerging countries, namely, Singapore, China, India, and South Korea are actively growing their biotech sectors, posing challenges to the long-standing US dominance.

04 Availability of Past Data

Firms in emerging economies are learning from the West's mistakes, avoiding costly infrastructure. The're innovatingwith lean, economical business models, enhancing global competitiveness.

So, What Do We Think?

BioTech Industry is NOT for Risk Averse Investors.

(Mostly New Investors and Investors solely for Profits and not Social Service)

- Approximately 50% of the BioTech Investments fail to make any profits AT ALL.
- Due to the hard-n-fast rules and regulations of almost all the countries around the globe on Biotech Innovations, even after successfully developing a product, it's very hard to get the approval from the concerned authorities.
- It only happens once in a blue moon that an investment catches multifold returns.

And, what can make THAT better?

Putting Up a Unified Front

One way of becoming more efficient is to become more collaborative. Firms need to give up on ego and join hands to make the industry survive as a whole. Survival always comes before Profits and Growth.

An Innovation Culture

In view of the investments and risks associated, firms need to be agile, willing to explore new ideas and open to insights produced outside their own walls. Senior management also need to encourage creative brainstorming.

Less R&D, More Profits

With a Business Model HEAVILY dependant on Venture Capital, Firms & StartUps should focus more on the profit side of the business in order to attract more investments. Money talks.

What did we NOT cover?

Just how BIG is BioTech?

BioTech in Tissue Engineering

Bio Informatics

BioTech in Industrial Processing

Cell Based Assays

BioTech in Latin America

PCR Tech and Chromatography

DNA Sequencing

Govt Initiatives in Biotech

BioTech in Fermentation

Nano BioTech

BioTech in MEA and India

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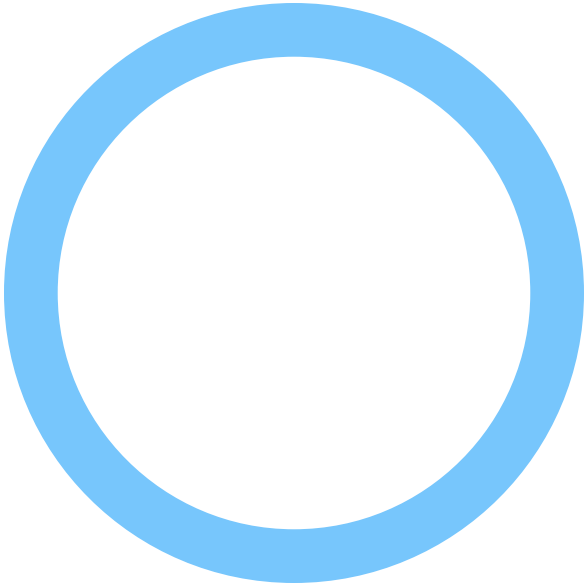
Appendix


McKinsey
Astrazeneca
Merck
Byjus
Pfizer
PwC
Publications

Our Consultants

Abhinav Uppal

Drowning in assignments, but staying afloat in style

Vansh Goyal

Finding it difficult to not breakdown crying

Vidit Juneja

Can't stop cribbing about PG Food